



DESIGN AND ANALYSIS OF ALGORITHMS

All Pairs Shortest Path (Floyd's Algorithm)

Reetinder Sidhu

Department of Computer Science and Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

All Pairs Shortest Path (Floyd's Algorithm)

Reetinder Sidhu

Department of Computer Science and
Engineering

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

UNIT 5: Limitations of Algorithmic Power and Coping with the Limitations



- Dynamic Programming
 - ▶ Computing a Binomial Coefficient
 - ▶ The Knapsack Problem
 - ▶ Memory Functions
 - ▶ **Warshall's and Floyd's Algorithms**
- Limitations of Algorithmic Power
 - ▶ Lower-Bound Arguments
 - ▶ Decision Trees
 - ▶ P, NP, and NP-Complete, NP-Hard Problems
- Coping with the Limitations
 - ▶ Backtracking
 - ▶ Branch-and-Bound. Architecture (microprocessor instruction set)

Concepts covered

- All Pairs Shortes Path (Floyd's Algorithm)
 - ▶ Definition
 - ▶ Algorithm
 - ▶ Example

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Problem Definition



- Given a undirected or directed graph, with weighted edges, find the shortest path between every pair of vertices
 - ▶ Dijkstra's algorithm found shortest paths from given vertex to remaining $n - 1$ vertices ($\Theta(n)$ paths)
 - ▶ Current problem is to find the shortest path between every pair of vertices ($\Theta(n^2)$ paths)
- Solution approach is similar to the transitive closure approach: Compute transitive closure via sequence of $n \times n$ matrices $R^{(0)}, \dots, R^{(k)}, \dots, R^{(n)}$ where $R^{(k)}[i, j] = 1$ iff there is nontrivial path from i to j with only first k vertices allowed as intermediate vertices
- Compute all pairs shortest paths via sequence of $n \times n$ matrices $D^{(0)}, \dots, D^{(k)}, \dots, D^{(n)}$ where $D^{(k)}[i, j]$ is the shortest path from i to j with only first k vertices allowed as intermediate vertices

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

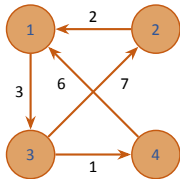
Example

- Example of all pairs shortest paths:

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example

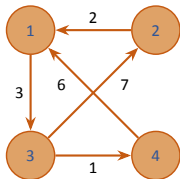
- Example of all pairs shortest paths:



ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example

- Example of all pairs shortest paths:

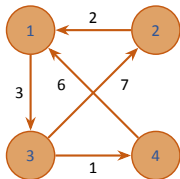


	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0

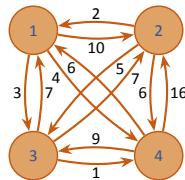
ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example

- Example of all pairs shortest paths:



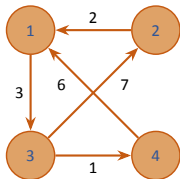
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



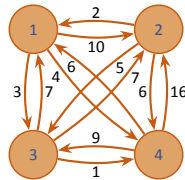
ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example

- Example of all pairs shortest paths:



	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	7	7	0	1
4	6	16	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Algorithm

Transitive Closure (Floyd's Algorithm)

```
1: procedure FLOYD( $A[1 \dots n, 1 \dots n]$ )
2:   ▷ Input: Weight matrix A of a graph with no negative length cycles
3:   ▷ Output: Distance matrix of shortest paths
4:    $D \leftarrow W$ 
5:   for  $k \leftarrow 1$  to  $n$  do
6:     for  $i \leftarrow 1$  to  $n$  do
7:       for  $j \leftarrow 1$  to  $n$  do
8:          $D[i, j] \leftarrow \min(D[i, j], D[i, k] + D[k, j])$ 
9:   return  $D$ 
```

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Algorithm

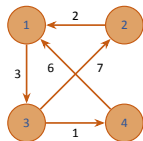
Transitive Closure (Floyd's Algorithm)

```
1: procedure FLOYD( $A[1 \dots n, 1 \dots n]$ )
2:   ▷ Input: Weight matrix A of a graph with no negative length cycles
3:   ▷ Output: Distance matrix of shortest paths
4:    $D \leftarrow W$ 
5:   for  $k \leftarrow 1$  to  $n$  do
6:     for  $i \leftarrow 1$  to  $n$  do
7:       for  $j \leftarrow 1$  to  $n$  do
8:          $D[i, j] \leftarrow \min(D[i, j], D[i, k] + D[k, j])$ 
9:   return  $D$ 
```

- Complexity: $\Theta(n^3)$

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

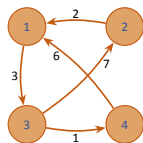
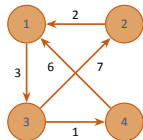
Example



	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example

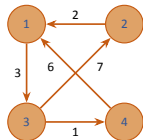


	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0

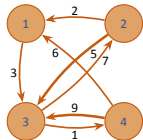
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Example



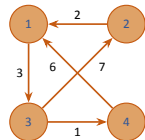
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



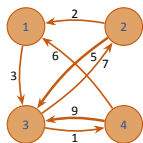
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

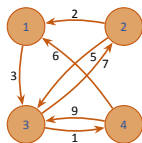
Example



	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



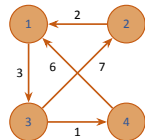
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



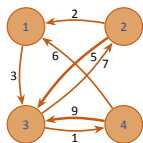
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

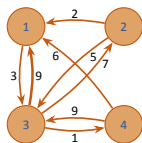
Example



	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



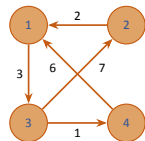
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



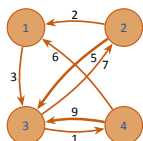
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

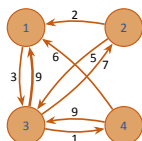
Example



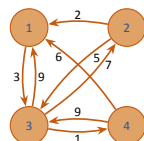
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



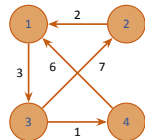
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0



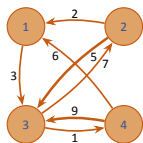
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

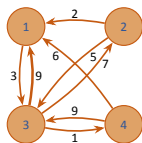
Example



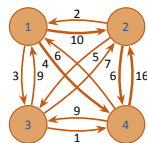
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



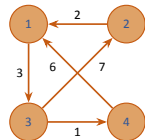
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0



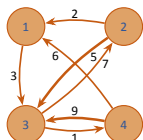
	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	9	7	0	1
4	6	16	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

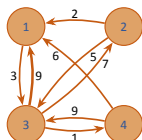
Example



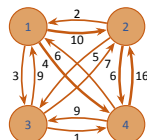
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



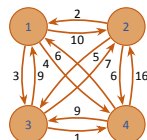
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0



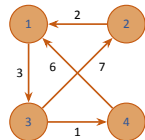
	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	9	7	0	1
4	6	16	9	0



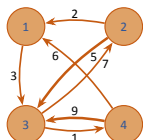
	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	9	7	0	1
4	6	16	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

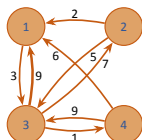
Example



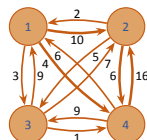
	1	2	3	4
1	0	∞	3	∞
2	2	0	∞	∞
3	∞	7	0	1
4	6	∞	∞	0



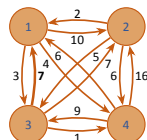
	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	∞	7	0	1
4	6	∞	9	0



	1	2	3	4
1	0	∞	3	∞
2	2	0	5	∞
3	9	7	0	1
4	6	∞	9	0



	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	9	7	0	1
4	6	16	9	0



	1	2	3	4
1	0	10	3	4
2	2	0	5	6
3	7	7	0	1
4	6	16	9	0

ALL PAIRS SHORTEST PATH (FLOYD'S ALGORITHM)

Think About It



- Give an example of a graph with negative weights for which Floyd's algorithm does not yield the correct result
- Enhance Floyd's algorithm so that shortest paths themselves, not just their lengths, can be found